MIL-C-46866 (MI)
23 March 1973
SUPERSEDING
MPD-11861B
13 July 1970

MILITARY SPECIFICATION

COMPOUND, EPOXY MOLDING AND APPLICATION OF

1. SCOPE

1.1 This specification covers one type of heat-curing epoxy compound suitable for transfer molding of delicate electronic assemblies.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-I-16923	Insulating Compound, Electrical, Embedding

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the Government, or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials

ASTM D150	Methods of Test for A-C Capacitance, Dielectric Constant, and Loss Characteristics of Electrical Insulating Materials

FSC 9330

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ASTM D256-72A	Impact Resistance of Plastics and Electrical Insulating Materials
ASTM D257	Methods of Test for Electrical Resistance of Insulating Materials
ASTM D618	Methods of Conditioning Plastics and Electrical Insulating Materials for Testing
ASTM D648	Method of Test for Deflection Temperature of Plastics Under Load
ASTM D790	Method of Test for Flexural Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet all the requirements of this specification.

3.2 Material. The epoxy molding compound shall be a granular form material.

3.3 Color. Unless otherwise specified, the material shall be black in color. The cured material may also have the following colors: brown, red, orange, yellow, green, blue or gray.

3.4 Curing. Curing of the epoxy molding compound shall be in accordance with the manufacturer's instructions. The cured material shall be in accordance with Table I.

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TABLE I - ELECTRICAL AND PHYSICAL PROPERTIES

Property	Test Method	Value
Dielectric strength (min)	4.3.1	350 volts/mil
Dielectric constant (max)	4.3.2	7.2 at 60 Hz 6.0 at 1 KHc 5.9 at 10 KHc
Dissipation factor (max)	4.3.2	0.040 at 60 Hz, 0.038 at 1 KHc 0.035 at 10 KHc
Volume resistivity (min)	4.3.3	8.6×10^{13} ohms/cm
Flexural strength (min)	4.3.4	10,000 psi
Impact strength (unnotched Izod) (min)	4.3.5	0.5 foot pounds
Heat distortion temperature (min)	4.3.6	325°F at 264 psi
Flow (min)	4.3.7	19 inches
Hardness, Shore D (min)	4.3.8	24

3.5 Shelf life. The minimum shelf life of the molding compound shall be six months from the date of the shipment.

3.6 Workmanship. The epoxy molding compound shall be uniform in quality and free from dirt and foreign material which might render the material unsuitable for its intended purpose.

3.7 Application of compound.

3.7.1 All electronic assemblies utilizing this material shall be completely encased with the epoxy molding compound. The thickness of the material shall be not less than 0.020 of an inch on any component or lead in the assembly.

3.7.2 The assembly being molded shall be clean and free of all contamination and shall be preheated to 245 ± 10°F prior to encapsulation. The molding pressure shall not exceed 700 pounds per square inch, and there may be 28/30 inches of mercury vacuum applied. The molding temperature shall be 250 ± 5°F with a cure time of 5+1, -0 minutes. The preforms shall be dielectrically heated to 200 ± 10°F.

3.7.3 Molded assemblies shall be cured completely without tackiness. There shall be no voids, blow holes, blisters, bubbles, broken or incomplete fills which would permit access of moisture to any surface of the molded assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. When required (see 6.2), preproduction inspection samples shall be tested prior to acceptance of any compound. Preproduction inspection shall consist of all the tests specified herein. A molded sample specimen 6" wide and 12" long and 1/8" thick molded at time manufactured items were produced will be submitted for all preproduction and production test evaluation samples.

4.3 Tests.

4.3.1 Dielectric strength. The test shall be made in accordance with Specification MIL-I-16923. Laboratory test conditions and test specimen conditioning shall be in accordance with ASTM D618, except the conditioning time shall be 24 hours in lieu of 48 hours. The short time test to breakdown of 500 volts a second shall be conducted.

4.3.2 Dielectric constant and dissipation factor. The tests shall be conducted in accordance with Specification MIL-I-16923 at the frequencies specified in table I, on three specimens 1/8 inch thick by two inches in diameter. Laboratory test conditions and specimen conditioning shall be in accordance with ASTM D618, except the conditioning time shall be 24 hours in lieu of 48 hours. A thin film of petrolatum shall be applied to the specimens with an aluminum foil electrode on each side of the specimen. The equipment for this test shall be as specified in ASTM D150, or equivalent.

4.3.3 Volume resistivity. The test shall be conducted in accordance with MIL-I-16923 on three specimens 1/8 inch thick by two inches in diameter at a temperature of 86°F maximum, except moisture conditioning shall be omitted. Laboratory test conditions and specimens conditioning shall be in accordance with ASTM D618, except the conditioning time shall be 24 hours in lieu of 48 hours. A d.c. potential of 500 volts shall be applied for one minute before each reading is recorded. Equipment used in this test shall be as specified in ASTM D257, or equivalent.

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4.3.4 Flexural strength. The test shall be conducted in accordance with ASTM D790. Five specimens shall be conditioned in accordance with ASTM D618, except the conditioning time shall be 24 hours in lieu of 48 hours.

4.3.5 Impact strength. The impact strength of the cured material shall be determined by ASTM D256.

4.3.6 Heat distortion temperature. The heat distortion temperature of the cured material shall be determined per ASTM D648.

4.3.7 Flow. The flow of the epoxy molding compound shall be determined using a Hull Spiral Flow mold (available from Hull Corporation, Hatboro, Pennsylvania), or equivalent. The material shall be prepared as follows:

(a) The molding compound shall be preformed at a pressure of 4380 ± 100 psi.

(b) The preforms shall be dielectrically preheated to an internal temperature of 200 ± 10°F.

(c) The Spiral mold temperature shall be 250 ± 5°F during the molding operation.

(d) Ram speed shall be so adjusted so as to accomplish injection of material into mold in 14 to 18 seconds as measured from time of positive contact with the material to the time of gelation.

(e) Cure time of the specimen shall be 3 1/2 minutes.

(f) Pressure at the transfer ram shall be 584 ± 25 psi.

A minimum of three specimens shall be used to determine the flow rate.

4.3.8 Hardness. Hardness of the cured epoxy molding compound shall be determined on a minimum of three specimens 1/8 inch thick by 2 to 3 inches in diameter, using a Shore "D" Hardness Instrument. The readings shall be taken on the specimens within three seconds after the press opens. Material shall be prepared as follows:

(a) Preform molding compound at a pressure of 4380 ± 100 psi.

(b) Dielectrically preheat preforms to a internal temperature of 200 ± 10°F.

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(c) Mold temperature shall be 250 ± 5°F.

(d) Ram speed shall be so adjusted so as to accomplish injection of material into mold in 14 to 18 seconds as measured from time of positive contact with the material to the time of gelation.

(e) Cure time of the specimen shall be 3 1/2 minutes.

(f) Pressure at the transfer ram shall be 584 ± 25 psi.

4.4 Acceptance inspection. Acceptance inspection shall be conducted on samples from each shipment or lot of manufacture to demonstrate quality of the material. Acceptance inspection shall consist of visual examination (see 4.5) and the following tests (see 4.3.2, 4.3.7 and 4.3.8):

(a) Dielectric constant and dissipation factor

(b) Flow

(c) Hardness

4.5 Visual inspection (cured). Each molded sample or production molded assembly shall be visually inspected to determine compliance with the requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be in accordance with MIL-P-14232, for Level C.

5.2 Marking for shipment. Each container shall be marked in a permanent manner with the following:

(a) Manufacturer's name and material designation

(b) Number of this specification

(c) Purchase order number

(d) Date of manufacture

(e) Date of shipment

(f) Date of expiration

(g) Batch or lot number

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6. NOTES

6.1 Intended use. The molding compound covered by this specification is intended to be used for transfer molding of micro-modules or standard electronic components and circuits to provide protection against humidity extremes, mechanical and thermal shock, and temperature extremes from -67°F to +185°F.

6.2 Ordering data. Procurement documents for the molding compound should specify the following.

(a) Title, number, and date of this specification.

(b) Quantity of preproduction samples required (see 4.2).

(c) Where preproduction testing is to be performed.

(d) Quantity of material required.

(e) Color, if other than black is desired.

6.3 Supersession data. This specification includes the requirements of Missile Purchase Description MPD-11861B dated 13 July 1970.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 9330-A577

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

DD, FORM 1426 JAN 66 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED

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